

A CHANGEPOINT ANALYSIS OF SOME DATA ARISING IN GAMMA-RAY ASTRONOMY

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Summary: Data consisting of arrival times of cosmic rays from the direction of a suspected periodic source are analysed. The objective is to detect a noisy periodic signal which may be present for part of the observation period. A standard approach in this type of problem is to reduce the arrival times modulo the expected signal period, hence obtaining points distributed on the circumference of a circle. In the absence of a periodic signal these points are uniformly distributed. A technique for detecting a change to non-uniformity is proposed and compared to one which attempts to identify changes in the arrival rate of cosmic rays.

1. INTRODUCTION

Celestial sources of pulsed radiation, commonly known as pulsars, have been the subject of intensive study by astrophysicists in recent years. Radio pulses from these sources can exhibit an amazing regularity. For instance, according to Rawley et al. (1987), "The stable rotation and sharp radio pulses of PSR 1937+21 makes this pulsar a clock whose long-term frequency stability approaches and may exceed that of the best atomic clocks". However, the sharpness of pulsed radiation can substantially diminish in the higher reaches (X-rays and gamma-rays) of the electromagnetic energy spectrum, with the result that the clocklike regularity may no longer be as evident as in the case of radio pulses; see Manchester & Taylor (1977, page 79) for an illustration of this. More recently, astrophysicists have been

searching for pulsars which emit VHE (very high energy) gamma-rays. Among data that are collected, those of particular interest are the arrival times of cosmic rays from the direction of a suspected gamma-ray source. Observations are usually made over a fixed length of time and can be thought of as arising from one of two components.

First, there is background radiation (noise) present throughout the observation period. Secondly, there may be one or more time segments during which the source is active in the sense that it emits detectable VHE gamma-rays. In astrophysical language these active segments are referred to as "bursts" or "transients". Astrophysicists still have difficulty in confirming the reality of such events because of the lack of appropriate statistical techniques and the unpredictable emission patterns of some suspected VHE gamma-ray sources.

While a pulsar's rotational period can be accurately determined from observations of its radio pulses, the arrival times of pulsed VHE gamma-rays often exhibit a measure of stochastic variation. This is a consequence of the diminished sharpness of the gamma-ray pulses. The arrival times $s_1, s_2, \dots$ of pulsed gamma-rays therefore have the structure

$$(1.1) \quad s_i = ip + \epsilon_i, \quad i \geq 1,$$

where p is the known pulsar period and $\epsilon_1, \epsilon_2, \dots$ is a sequence of independent, identically distributed, random variables about which one knows only that they are non-uniformly distributed over the interval $(0, p)$. In reality, however, only a small fraction of pulsed gamma-rays are observed. This is primarily because of the weakness of pulses and the limited detection capability of the receiving apparatus. The model (1.1) must therefore be replaced by

$$(1.2) \quad s_i = k_i p + \epsilon_i, \quad i \geq 1,$$

where $k_i, i \geq 1$, is a sequence of random positive integers with expectation substantially larger than 1, and where the ϵ_i are as before. In point process terminology, (1.2) is a thinned version of (1.1).

The noise component supposedly generates arrival times $\eta_1, \eta_2, \dots$ in a Poisson process with unknown intensity $\nu > 0$. That is,

$$(1.3) \quad \eta_i = e_1 + \dots + e_i, \quad i \geq 1$$

where the interarrival times $e_1, e_2, \dots$ are independently and exponentially distributed with common mean ν^{-1} . In our particular application ν^{-1} is substantially larger than p , and this turns out to be a crucial point in the ensuing analysis.

Thus, a typical data set consists of a sequence of arrival times $\{t_i\}$, each arrival time representing either noise or pulsed radiation. Of course, one does not know which of these components any given arrival time represents. The observed sequence of arrival times is the superposition of the pulsed (when it is detectable) and noise (always present) processes.

The main purpose of our analysis is to detect the presence of a pulsed component at a specified period and to identify the stretch of data in which this component is present. This stretch of data can then be further analysed to gain information regarding the pulse profile of the radiation. A more general problem, which will not be addressed here, is to detect the presence of a signal of unknown period. If all arriving gamma-rays could be observed, the presence of an almost periodic component such as s_i in (1.1) could be detected quite easily. However, as pointed out earlier, only a small fraction of arriving pulsed radiation is actually observed, so that the signal s_i in (1.2) is not periodic in the strict sense. Thus, during an active (pulsed radiation plus noise) segment no "obvious" periodic component could be evident in the data. The principle is illustrated in figure 1 using some simulated data. Figure 1(a) gives arrival times simulated from (1.2) where $p = 1$, k_i is a random integer between 1 and 50 and $\epsilon_i = 1/2$ with probability one. Figure 1(b) gives noise arrival times simulated from (1.3) with mean interarrival time $\nu^{-1} = 5$. Figure 1(c) is the superposition of Figures 1(a) and 1(b) and represents the observed arrival time process. Note the apparent lack of periodicity in this process.

Because of the intrinsically periodic nature of pulsar emission, astrophysicists routinely "fold" the observed arrival times t_i with respect to the pulsar period p , thereby obtaining observations θ_i on a circle of unit radius:

$$(1.4) \quad \theta_i = 2\pi(t_i/p - \text{integer part of } t_i/p).$$

Figure 2 shows the result of folding figure 1(c) in accordance with (1.4). The "hidden" periodicity is now quite obvious.

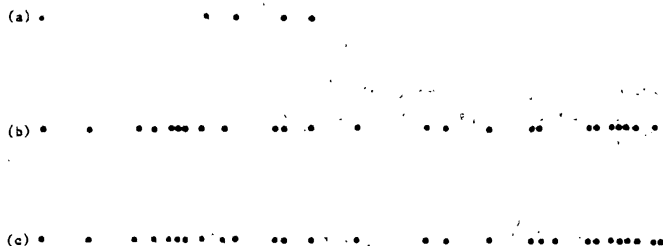


Figure 1 (a) Simulated gamma-ray arrival times
(b) Simulated noise arrival times
(c) Observed arrival times: superposition of (a) and (b)

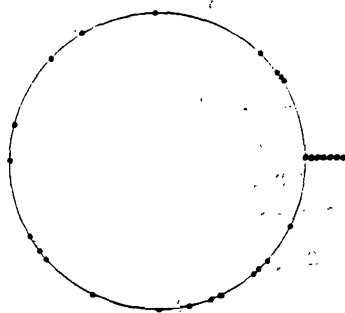


Figure 2. Folded arrival times from figure 1(c)

The data to be analysed in this paper consist of 1554 interarrival times of VHE cosmic rays from the direction of the pulsar PSR 1802-23. (These data are available from the author upon request.) In section 2 it is shown that no change in arrival rate is apparent in the data. On the other hand, an analysis of folded arrival times in section 3 suggests the presence of a periodic signal in one segment of the data. Some of the more technical aspects of the analysis are discussed in an appendix.

2. ANALYSIS OF ARRIVAL RATE

Raubenheimer, et al. (1986) gave $p = 0,11255$ seconds as the period of PSR 1802-23. The average noise interarrival time is expected to be of the order of $10p$ seconds. Thus, it would be natural to expect the onset of pulsed radiation to lead to a substantial decrease in the average interarrival time. Recall from section 1, however, that only a small proportion of any pulsed radiation is actually detected, so that no marked decrease in the average interarrival time may be noticeable.

The average interarrival time over the full set of observations is 1,388 seconds. Figure 3 shows a Q-Q plot, assuming an underlying exponential distribution. There is no marked deviation from the expected line. This is already an indication that the mean interarrival time is not changing. The data were also checked for autocorrelation between interarrival times, but no values significantly different from zero were found.

More formally, an analysis similar in spirit to one proposed by Cox & Lewis (1966, page 39) can be used. The data are split into 155 consecutive non-overlapping groups of 10 observations each, ignoring for this purpose the first four observations. Then

$$x_i = (9.5)^{\frac{1}{2}} \log_e (\text{total in group } i)$$

is calculated for $i = 1, \dots, 155$. Under an assumption of exponentially distributed interarrival times the x_i are independent and approximately normally distributed with unit variance. The sample variance of the 155 x_i values is, in fact, 0,983. A cumulative sum chart, which plots

$$C_n = (1/155)^{1/2} \sum_{i=1}^n (x_i - \bar{x}) \text{ against } n \text{ for } n = 1, \dots, 154$$

is shown in figure 4. The average squared deviation is

$$(1/154) \sum_{n=1}^{154} C_n^2 = 0,127$$

which corresponds to a (large sample) significance probability of about 0,43; see Lombard (1989, page 12, equation (3.19)). Similar non-significant results were obtained when the group size was increased or decreased. Thus, there is no evidence from these analyses that the observed average interarrival times have changed by a statistically significant amount.

The preceding analysis ignores the intrinsically periodic nature of pulsar emission. To take account of this, the arrival times can be folded as in (1.3) yielding observations, θ , on the circumference of a circle. Then, if the mean interarrival time in the Poisson noise process is large compared to the pulsar period p , it can be shown that the corresponding θ 's are approximately independent and uniformly distributed. On the other hand, those θ 's corresponding to pulsed radiation will exhibit a non-uniform distribution, namely that of $2\pi\epsilon$ for the ϵ in (1.1). In this way a change from a uniform to a non-uniform distribution of θ 's corresponds to the onset of pulsed radiation; see appendix 1.

3. ANALYSIS OF THE FOLDED ARRIVAL TIMES

Let $\theta_1, \dots, \theta_n$ be independently distributed points on the circumference of a circle of unit radius. These can be viewed as the end points of unit vectors extending from the centre of the circle. Each unit vector can, in turn, be identified with its polar angle, measured anti-clockwise from an arbitrary, but fixed, reference vector; see Mardia (1972, pp. 18-19). Denote the resultant length of the n unit vectors by R_n . In terms of polar angles,

$$(3.1) \quad R_n^2 = \left[\sum_{i=1}^n \cos \theta_i \right]^2 + \left[\sum_{i=1}^n \sin \theta_i \right]^2$$

If the underlying distribution is uniform on the circle, R_n^2 can be expected to be small because of the cancellation of unit vectors pointing in opposite directions. If, however, the distribution is concentrated in some particular direction little cancellation takes place and a large value of R_n^2 can be expected. A well-known test of circular uniformity is based on the Rayleigh statistic $2n^{-1}R_n^2$ which, for large n , has approximately a chi-square distribution with two degrees of freedom under the hypothesis of circular uniformity; see Mardia (1972, pp. 20 and 133). Table 1 gives the value of the Rayleigh statistic for successive non-overlapping groups of 200 folded arrival times, the last group consisting of 154 observations. The significance probabilities in the fourth and fifth groups strongly suggest the presence of non-uniformly distributed data. The overall significance level of the eight groups combined is defined to be the probability that at least one of eight independent chi-squared (2 df) random variables will exceed the value 9,627. This is $1-(1-0.008)^8 = 0,062$.

Table 1. Rayleigh statistic for successive groups of 200 observations

Observations	Rayleigh ^a statistic	Significance ^b probability
1 to 200	0,338	0,844
201 to 400	1,557	0,459
401 to 600	0,407	0,816
601 to 800	9,627	0,008
801 to 1000	8,418	0,015
1001 to 1200	1,345	0,510
1201 to 1400	5,443	0,066
1401 to 1554	3,599	0,273

^a $2n^{-1}R_n^2$ with R_n given in (3.1)

^b $\text{Prob}(2n^{-1}R_n^2 > x) \approx e^{-x/2}$

A more informative procedure, which is not dependent on an arbitrary grouping of the data, is to plot R_n^2 against n for $n = 1, 2, \dots$; see figure 5.

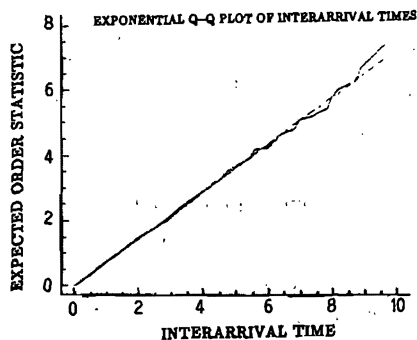


Figure 3. Exponential Q-Q plot of interarrival times

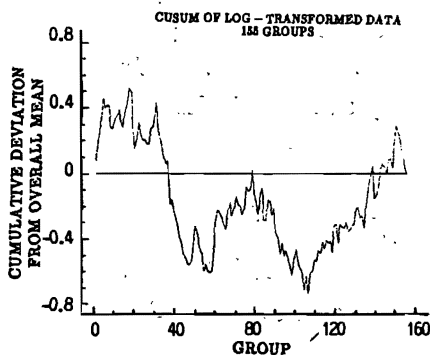


Figure 4. Cusum of log-transformed data 155 groups

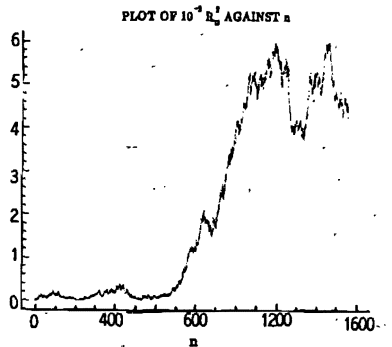


Figure 5. Plot of $10^{-3} R_n^2$ against n

The strong upward drift between the (approximately) 600th and 1100th observations is indicative of a change from uniformly to non-uniformly distributed data. A weighted average of the Rayleigh statistic over N observations is

$$(3.2) \quad \bar{R}_N^2 = \sum_{n=1}^N (2n/N^2)(2R_n^2/n) = 4N^{-2} \sum_{n=1}^N R_n^2$$

and for our particular data set ($N = 1554$) the observed value of $\bar{R}_N^2$ is 5,516. This corresponds to a significance level of 0,042; see appendix 2. Thus there is some evidence that pulsed radiation (at the given period $p = 0,11255$ seconds) is present during part of the observation time.

More precise estimates of the points of change can be obtained by adapting a technique proposed by Lombard (1986, section 4.3). For this, set

$$\bar{C}_m = m^{-1} \sum_{j=1}^m \cos \theta_j; \quad \bar{C}^{N-m} = (N-m)^{-1} \sum_{j=m+1}^N \cos \theta_j$$

for $m = 1, \dots, N-1$ and define $\bar{S}_m$ and $\bar{S}^{N-m}$ similarly by replacing $\cos \theta_i$ by $\sin \theta_i$. Note that the vectors

$$(\bar{C}_m, \bar{S}_m) \text{ and } (\bar{C}^{N-m}, \bar{S}^{N-m})$$

represent the means of the vectors corresponding to the observations $\theta_1, \dots, \theta_m$ and $\theta_{m+1}, \dots, \theta_N$ respectively. For the full data set ($N = 1554$) the squared distance

$$(\bar{C}_m - \bar{C}^{N-m})^2 + (\bar{S}_m - \bar{S}^{N-m})^2$$

between these means is a maximum at $m = 1059$ which serves as an estimate of a change point.

We now delete the last 495 ($= 1554 - 1059$) observations and repeat the above calculation on the remaining $N = 1059$. This gives $m = 677$ as estimate of the second change point.

The analysis thus far suggests that the first 677 and last 495 arrivals are generated by a pure noise process while the remaining 382 contain a noisy component at period 0,11255 seconds.

Some confirmation of an entirely different nature follows from a consideration of elementary estimates of the probability density functions which supposedly generate the folded data. Since we are dealing with circle data, the orthogonal series estimator is particularly well suited to density estimation; see Silverman (1986, page 23). The density, f , can be expanded in a Fourier series

$$f(\theta) = \pi/2 + \sum_{\nu \geq 1} f_{\nu} \phi_{\nu}(\theta), \quad 0 \leq \theta < 2\pi,$$

where

$$\phi_{2n-1}(\theta) = \pi^{-\frac{1}{2}} \cos(n\theta)$$

$$\phi_{2n}(\theta) = \pi^{-\frac{1}{2}} \sin(n\theta); \quad n \geq 1,$$

and

$$f_{\nu} = \int_0^{2\pi} f(\theta) \phi_{\nu}(\theta) d\theta.$$

Then

$$\hat{f}_{\nu} = m^{-1} \sum_{i=1}^m \phi_{\nu}(\theta_i)$$

is an unbiased estimator of f_{ν} based on data $\theta_1, \dots, \theta_m$. Under an hypothesis of uniformity, $H_0: f_1 = f_2 = \dots = 0$, we have that $(2\pi m)^{\frac{1}{2}} \hat{f}_{\nu}$, $\nu \geq 1$, are approximately independent and normal $(0, 1)$ for large m . Restricting attention to the first K coefficients, where K is a large positive integer,

$$M = m^{\frac{1}{2}} \max(|\hat{f}_{\nu}|; 1 \leq \nu \leq K)$$

can be used to test H_0 . For $K = 100$ we have

$$(3.3) \quad P[M > b | H_0] \approx 1 - (2\Phi(b) - 1)^{100}$$

with Φ denoting the normal $(0, 1)$ distribution function.

For the subset of data consisting of the first 600 and last 454 folded arrival times we find $M = 3,184$. Now $P[M > 3,184 | H_0] \approx 0,14$ so that there is no reason to reject H_0 . On the other hand, the data $\theta_{700}, \dots, \theta_{1000}$ gives, with $m = 301$,

$$(2\pi m)^{\frac{1}{2}} \hat{f}_2 = -4,077 \text{ and } (2\pi m)^{\frac{1}{2}} |\hat{f}_\nu| \leq 3,3 \text{ for } \nu \neq 2.$$

Figure 6 shows a normal probability plot of the 100 estimated coefficients on which the one abberant observation can be clearly seen. Since

$$P[M \geq 4,077 | H_0] \approx 0,005$$

we reject the uniformity hypothesis for these data. Figure 7 shows plots of the empirical distribution functions of the two subsets of the data together with the cdf of a uniform distribution.

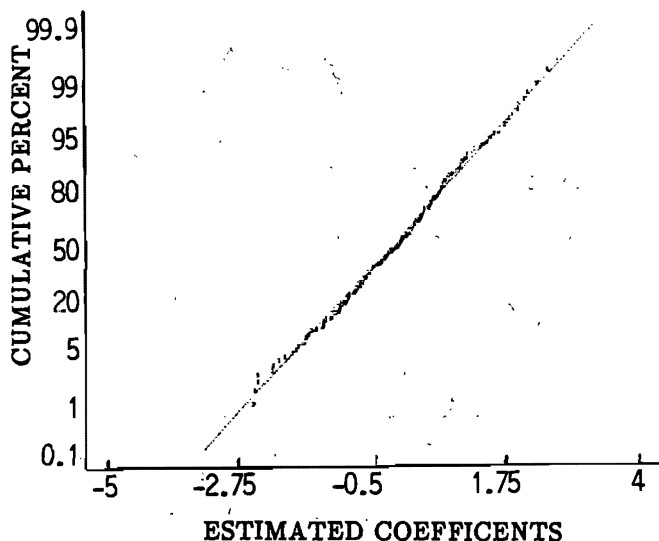
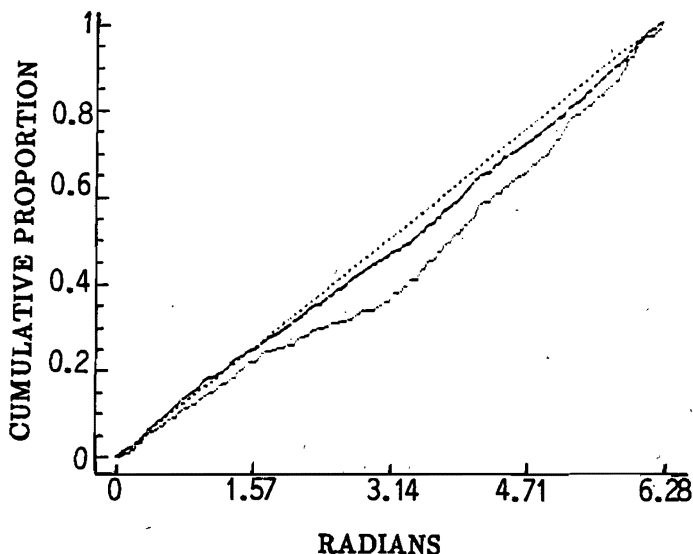


Figure 6. Normal probability plot of 100 estimated Fourier coefficients



dotted line: Uniform cdf

dark jagged curve: observations 1 - 600 and 7700 - 1554[†]

lighter jagged curve: observations 700 - 1000

[†] The statistical analysis suggests that the apparent deviation from uniformity is not significant.

Figure 7. Empirical CDFs of folded data

4. MAIN CONCLUSIONS

An analysis of cosmic ray arrival times has been given. While no marked change in arrival rate is evident, analysis of the folded data suggests the presence of a noisy periodic signal in part of the data. A significance test, based on a weighted average of Rayleigh statistics, has been derived. The procedure is generally applicable to detecting changes from uniformity in time-ordered circle data.

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REFERENCES

- BILLINGSLEY, P. (1968). *Convergence of probability measures*. New York: John Wiley and Sons.
- COX, D.R., and LEWIS, P.A.W. (1966). *The statistical analysis of series of events*. London: Methuen and Co.
- LOMBARD, F. (1986). The change-point problem for angular data: A nonparametric approach. *Technometrics*, 28, 391–397.
- LOMBARD, F. (1989). Some recent developments in the analysis of changepoint data. *South African Statist. J.*, 23, 1–21.
- MANCHESTER, R.N., and TAYLOR, J.H. (1977). *Pulsars*. San Francisco: W.H. Freeman and Co.
- MARDIA, K.V. (1972). *Statistics of directional data*. New York: Academic Press.
- RAUBENHEIMER, B.C., NORTH, A.R., DE JAGER, O.C., VAN URK, G. and VAN TONDER, A.J. (1986). Pulsed VHE gamma-rays from the COS-B source 2CG006–00. *Astrophysical J. (Letters)*, 307, L143–L147.
- RAWLEY, L.A., TAYLOR, J.H., DAVIS, M.M. and ALLAN, D.W. (1987). Millisecond pulsar PSR 1937 + 21: A highly stable clock. *Science*, 238, 761–765.
- ROTHMAN, E.D. and WOODROOFE, M. (1972). A Cramér-Von Mises type statistic for testing symmetry. *Ann. Math. Statist.*, 43, 2035–2038.
- SILVERMAN, B.W. (1986). *Density Estimation for Statistics and Data Analysis*. London: Chapman and Hall.

APPENDIX

1. DISTRIBUTION OF FOLDED ARRIVAL TIMES

For any real random variable x , let $f(x)$ denote the result when x is folded around a circle of circumference p :

$$f(x) = 2\pi((x/p) - \text{integer part of } (x/p)).$$

Recall the dual interpretations of $f(x)$ as the endpoint of a unit vector extending from the origin and as the polar angle of this vector. Observe that, with the second interpretation,

$$f(x_1 + x_2) = f(f(x_1) + f(x_2)).$$

The following two results will be required below.

Poincaré's Theorem (Mardia (1972, p. 91)).

Let x be a continuous random variable with standard deviation σ . If $\sigma/p \rightarrow \infty$, then the distribution of $f(x)$ approaches the uniform distribution on the circle.

Lemma

If $u_1, u_2, \dots$ are independently and uniformly distributed on the circle, then so are $f(u_1), f(u_1 + u_2), \dots$.

Proof

It is well known (Mardia (1972, pp. 50-51)) that $f(u_1), f(u_1 + u_2), \dots$ are all uniformly distributed. As to their independence, observe that, in view of the independence of u_1 and u_2 , the conditional distribution of $f(u_1 + u_2) = f(f(u_1) + f(u_2))$, given $f(u_1) = \theta$, is the same as the unconditional distribution of $f(\theta + f(u_2))$. But $f(\theta + f(u_2))$ is obtained by anti-clockwise rotation of the vector $f(u_2)$ through an angle θ and such a rotation obviously preserves the uniform distribution of $f(u_2)$. Thus, the conditional

distribution of $f(u_1 + u_2)$, given $f(u_1)$, is the same as its unconditional distribution, namely uniform. Hence $f(u_1)$ and $f(u_1 + u_2)$ are independently distributed. The required result follows in full generality by an obvious extension of the above argument. ■

Because the e_i , $i \geq 1$, in (1.3) are independently distributed, so are $u_i = f(e_i)$, $i \geq 1$. We also assumed that ν^{-1} , the common standard deviation of the e_i , was much larger than p . Then, by Poincaré's theorem, the u_i are also (approximately) uniformly distributed. Keeping (1.3) in mind, it then follows from the above lemma that the folded noise arrival times $f(\eta_i) = f(u_1 + \dots + u_i)$, $i \geq 1$, are (approximately) independently and uniformly distributed.

Finally, it follows from (1.2) and (1.3) that the folded pulsed arrival times are $f(s_i) = 2\pi\epsilon_i/p$. These are independent because, by assumption, the ϵ_i are so. During an active period of the pulsar an observed arrival time t_i is an s_i with (unknown) probability q and an η_i with probability $1-q$. Thus, the distribution of $f(t_i)$ is a mixture of the non-uniform distribution of $f(s_i)$ and the (approximately) uniform distribution of $f(\eta_i)$.

It should be noted that the analysis in terms of folded arrival times is equivalent to a consideration of the Fourier transform and periodogram of the point process of arrival times. The arrival times $\{t_i\}_{i=1}^n$ specify a realisation of part of a point process $\{P(t), t \geq 0\}$ which can be represented in the form

$$P(dt) = \sum_{i=1}^n \delta(t-t_i)dt$$

where $\delta(\cdot)$ denotes the Dirac delta function. The Fourier transform of $P(\cdot)$ at frequency $\nu = 1/p$ is

$$F_n(p) = \int_0^\infty \exp(\sqrt{-1} 2\pi t/p) P(dt) = \sum_{i=1}^n \exp(\sqrt{-1} 2\pi t_i/p) = \sum_{i=1}^n \exp(\sqrt{-1} \theta_i),$$

with θ_i given in (1.4), and the periodogram at period $1/p$ is

$$2n^{-1} |F_n(p)|^2 = 2n^{-1} R_n^2,$$

with R_n^2 given in (3.1.)

2. ASYMPTOTIC DISTRIBUTION OF $\bar{R}_N^2$ IN (3.2)

Let the data $\theta_1, \dots, \theta_N$ be independently and uniformly distributed on $[0, 2\pi]$. Then the random variables $\{\cos \theta_i, \sin \theta_j; 1 \leq i, j \leq N\}$ are mutually independent with means zero and variances $1/2$. Set

$$C_N(t) = (2/N)^{1/2} \sum_{j=1}^N \cos \theta_j, \quad S_N(t) = (2/N)^{1/2} \sum_{j=1}^N \sin \theta_j$$

where both sums are over $j = 1, \dots, \text{int}((N+1)t)$ and int denotes "integer part". Observe that

$$\bar{R}_N^2 = 2N^{-1}(N+1) \int_0^1 [C_N^2(t) + S_N^2(t)] dt.$$

From Donsker's theorem (Billingsley (1968, page 137)) we have $C_N(\cdot) \xrightarrow{\mathcal{D}} B_1(\cdot)$ and $S_N(\cdot) \xrightarrow{\mathcal{D}} B_2(\cdot)$ where $\xrightarrow{\mathcal{D}}$ denotes convergence in distribution and B_1 and B_2 are independent standard Brownian motions. This fact, together with the continuous mapping theorem (Billingsley (1968, page 30)) implies that the distribution of $\bar{R}_N^2$ is for large N approximately that of

$$Q = 2 \int_0^1 (B_1^2(t) + B_2^2(t)) dt.$$

An easy consequence of corollary 2 of Rothman and Woodroffe (1972) is that

$$(A2.1) \quad Q = 4 \sum_{k \geq 1} ((k-1/2)\pi)^{-2} e_k$$

where $e_1, e_2, \dots$ are independent variates with a common negative exponential distribution: $P(e > y) = \exp(-y)$ for $y > 0$. A representation of the distribution function of Q in the form of an infinite series can be found using the method indicated by Rothman and Woodroffe. Alternatively, the distribution of Q is approximated to a high degree of accuracy by that of

$$\begin{aligned} Q^* &= (4/\pi) e_1 + (4/\pi)^2 \sum_{k \geq 2} (2k-1)^{-1} \\ &= 1.62 e_1 + 0.38 \end{aligned}$$

which consists of the first term in the representation (A2.1) of Q plus a constant, 0.38, which makes

$$E(Q^*) = E(Q) = 2.$$

Then

$$P[Q > b] \approx \exp((0.38 - b)/1.62).$$